Unit 2: Study of Population & Migration

Day 18, Unit 2, October 4 & 5

Agenda

1. **Turn in HDI Project**
2. **Unit 2 Important Dates**
3. **Notes - 2.1 & 2.2 Population Distribution**
4. **AP Classroom Questions & Enrichment: Population Trends**

Unit 2 Important Dates

- **Agendas Due: Friday, 10/13**
- **Population Project Due: Tuesday, 10/24 (A) & Wednesday, 10/25 (B)**
- **Chapters 2 & 3 Reading Quiz & Guides Due: Friday, 11/3 (A) & Monday, 11/6 (B)**
- **Unit 2 Exam: Tuesday, 11/7 (A) and Wednesday, 11/8 (B)**

Recommended Pacing/Chunking of Chapters 2 & 3

A Day:

Ch. 2 pages 29-46 (FN, KQ1, KQ2) by Wed. 10/11

Ch. 2 pages 46-58 (KQ3, KQ4, KQ5) by Wed. 10/18

Ch. 3 pages 60-73 (FN, KQ1, KQ2) by Tues. 10/24

Ch. 3 pages 73-88 (KQ3, KQ4) by Tues. 10/31

Study for reading quiz 11/1-11/2

B Day:

Ch. 2 pages 29-46 (FN, KQ1, KQ2) by Thurs. 10/12

Ch. 2 pages 46-58 (KQ3, KQ4, KQ5) by Thurs. 10/19

Ch. 3 pages 60-73 (FN, KQ1, KQ2) by Wed. 10/25

Ch. 3 pages 73-88 (KQ3, KQ4) by Friday, 11/3

Study for reading quiz 11/4-11/5

Objectives

2.1 Population Distribution

- Identify the factors that influence the distribution of human populations at different scales.
- Define methods geographers use to calculate population density.
- Explain the differences between and the impact of methods used to calculate population density.

2.1: Consequences of Population Distribution

- Explain how population distribution and density affect society.

Record notes from video

3 D’s of Population

- Demographics
- Distribution
- Density

Demography

- **Study of the composition of human populations**
- Demographics
  - Quantifiable statistics that measure characteristics of a given population
    - Gender/Sex
    - Age groups
    - Ethnicity
    - Income levels
    - Migrant populations
- Demographers analyze variables within certain populations and their effects

Demographic Indicators

- CBR = crude birth rate = # of births in a given year per 1,000 people
- CDR = crude death rate = # of deaths in a given year per 1,000 people
- TFR = total fertility rate = average number of children one woman in a given country will have during her child-bearing years (15 - 49) ➜ pop. replacement level = 2.1
- Life Expectancy = average # of years a person is expected to live (check this out)
- IMR = infant mortality rate = # of deaths of children under the age of 1 per 1,000 live births

Distribution

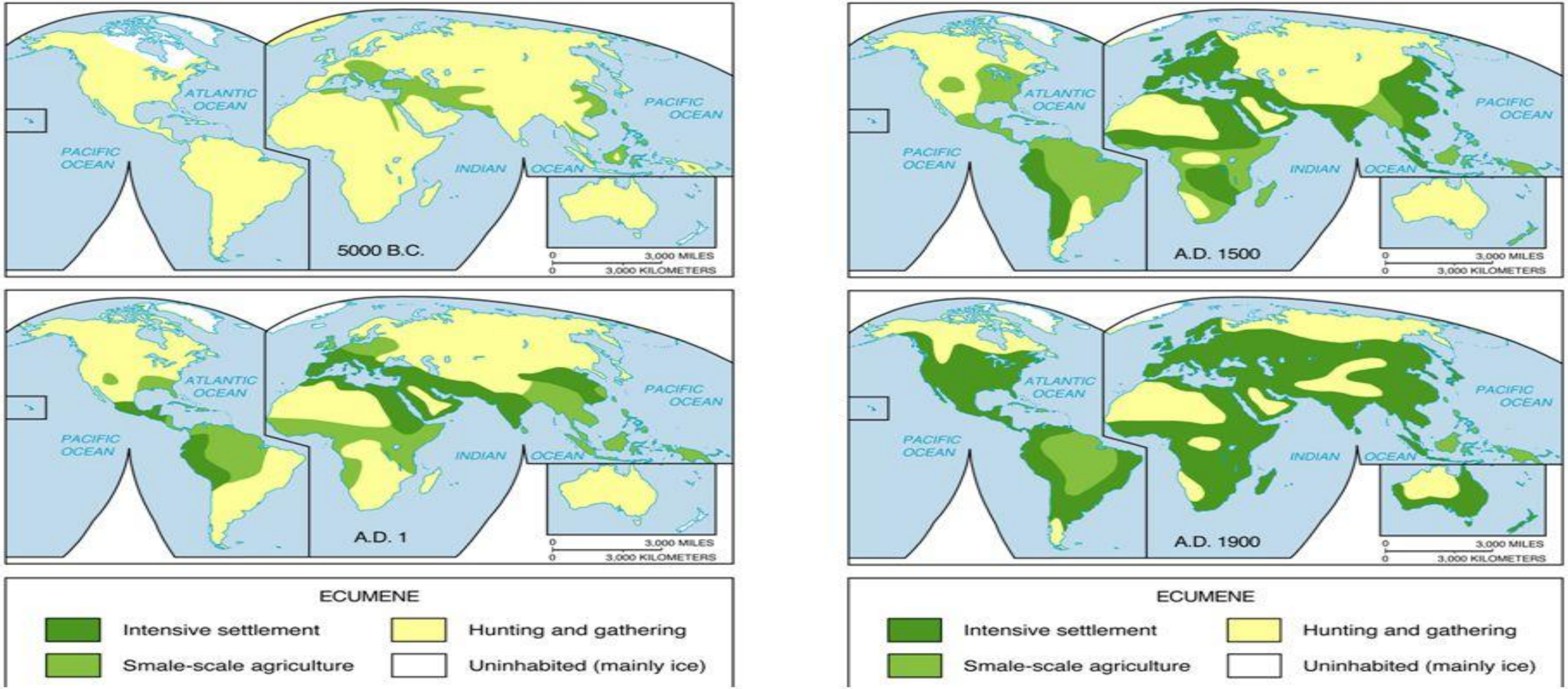
- **The pattern of where people live**
- **World distribution is uneven**
  - Places which are **sparsely** populated contain few people
    - Tend to be harsher environments
  - Places which are **densely** populated contain many people
    - Tend to be clustered around water sources and arable land

Ecumene

- The portion of the Earth’s surface occupied by permanent human settlements.
- *In fact, about 75 percent of the population of the world live on 5 percent of the Earth’s surface. Less than 30 percent of the earth’s surface is land area, but large parts of that land cannot support any substantial number of inhabitants.*

Expansion of the Ecumene

5000 B.C.E.– 1900 C.E.



Density

- Measure of total population relative to land size
- Population density is higher in areas with more people and less land
- Countries with high (arithmetic) population density:
    - India
    - Pakistan
    - China
    - Nigeria
    - Kenya
    - Indonesia

$$\textbf{Population density} = \frac{\textbf{Total population}}{\textbf{Surface km}^2} = \textbf{people/km}^2$$

High = over 100 people/km^2
Medium = between 50 and 100 people/km^2
Low = between 25 and 50 people/km^2
Very low = under 25 people/km^2

Arithmetic Population Density

- Population of a region or country expressed as an average per unit of area
- Allows general comparisons of the number of people living in different regions
- Doesn't really tell us anything else

Population ÷ Total Land

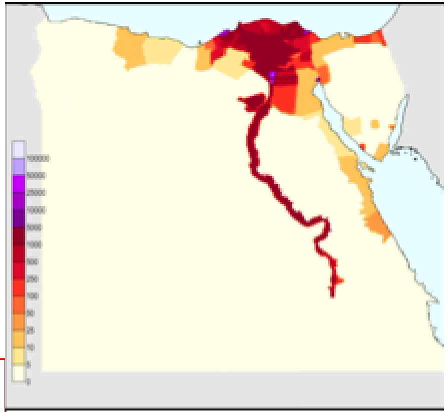
- Population of Egypt = 99,413,317 divided by
- Land = 995,450 sq km

Arithmetic Density = ~100 people per sq km

Physiological Population Density

Population ÷ Total Arable Land

- Is more representative of how crowded an area is
  - People do not tend to live in non-arable land
- Will "the capacity of the land yield enough food for the needs of the people?"



- Population of Egypt = 99,413,317
- Arable land = 2.8% of 995,450 sq km = 27,872

Physiological Density = 99,413,317/27,872 = ~3567 people per sq km

Agricultural Population Density

Farmers ÷ Arable Land

- Shows economic differences
- High ag density = heavy pressure on land for food = less technology and less ag investment, tend to be less developed (need a lot a farmers and it still might not be enough)
- Low ag density = can get more food with less farmers = may tend to import food = tend to be more developed - more farming tech and $$ for tech

- Egyptian Farmers make up 25.8% of labor force (pop = 99,413,317.258) = 25,648,635
- Arable land = 2.8% of 995,450 sq km = 27,872
- 25,648,635 / 27,872 =

Agricultural Density = 920 farmers per sq km

Where do we live? Take notes from video.

2.2

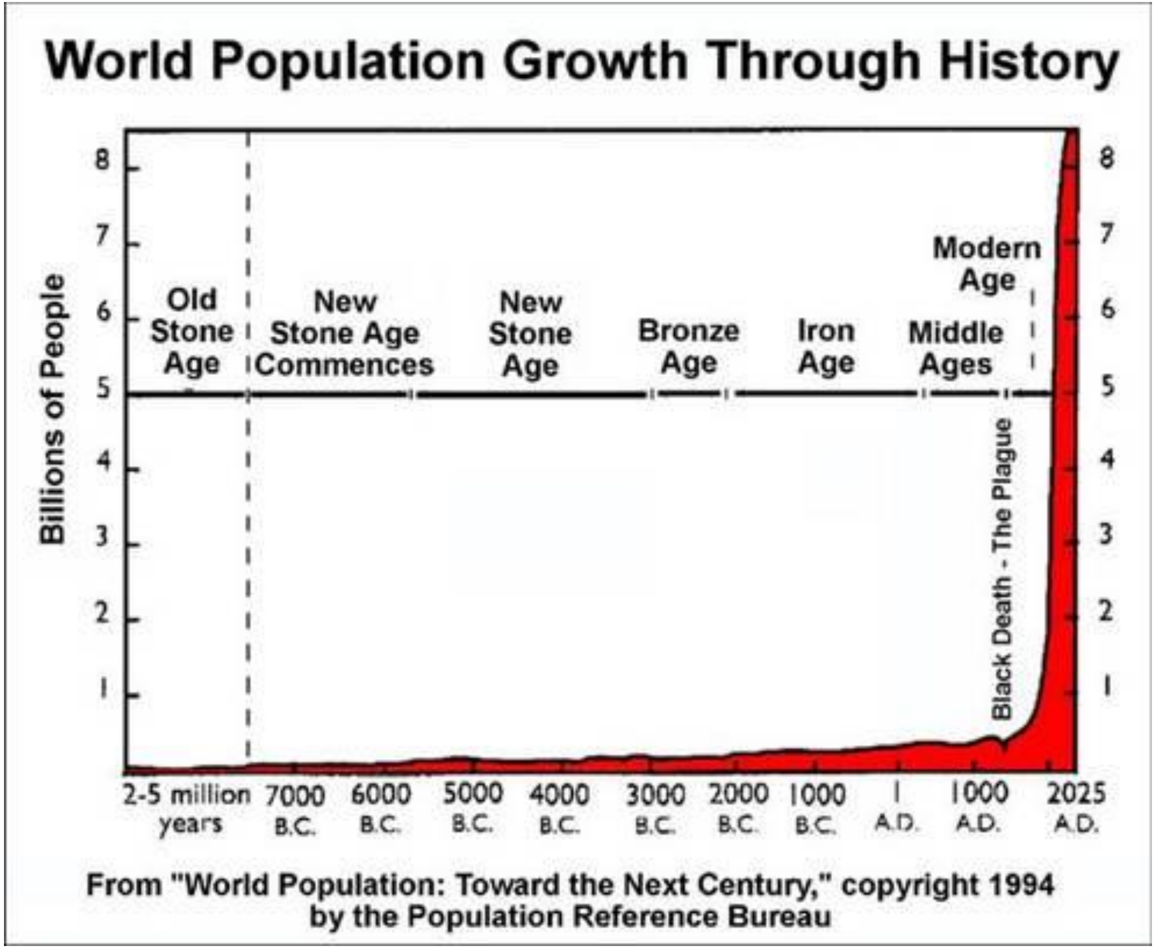
Consequences of Population Distribution

Trends: J-Curve

EXPONENTIAL GROWTH

2. 4. 8. 16. 32

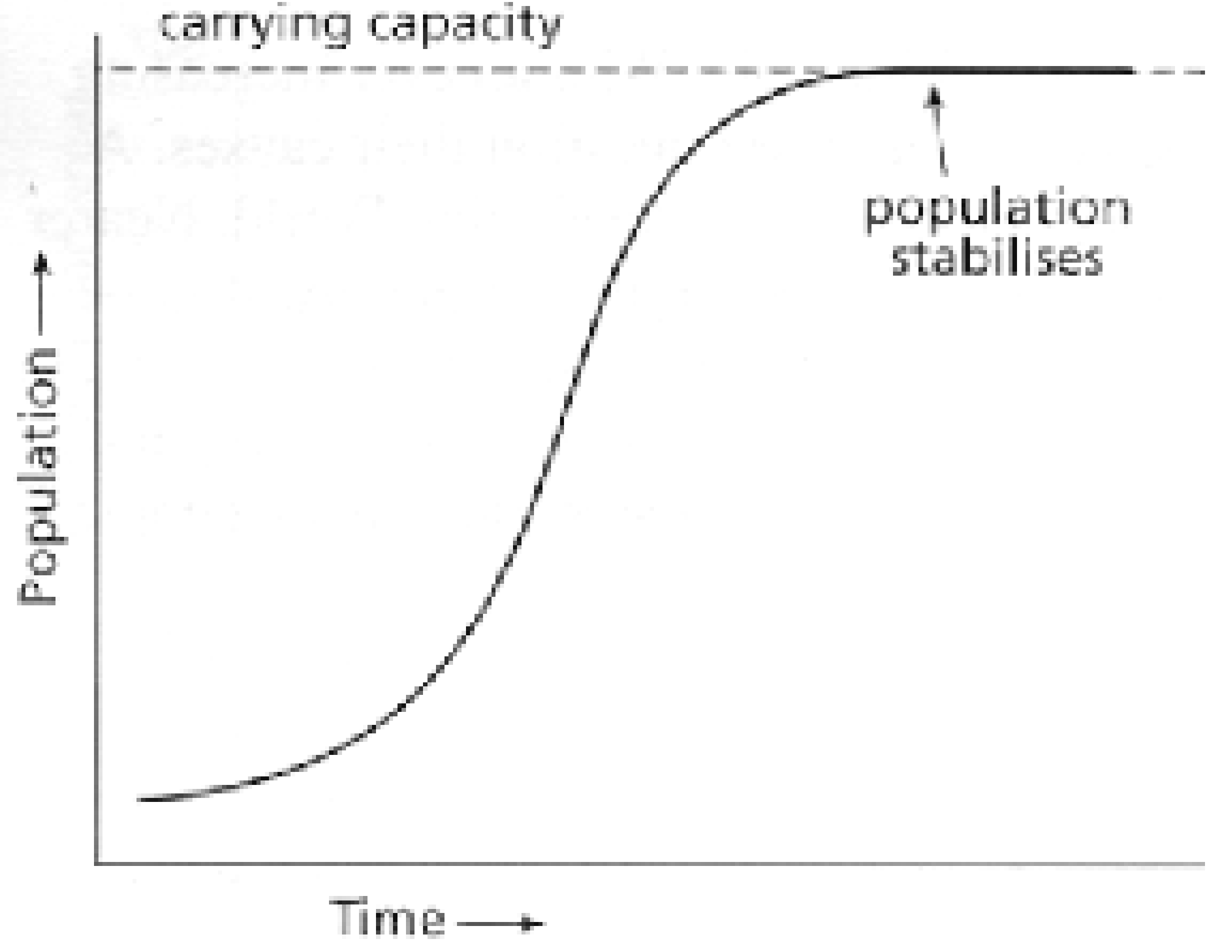
Once children are born, they grow up and have their own children...



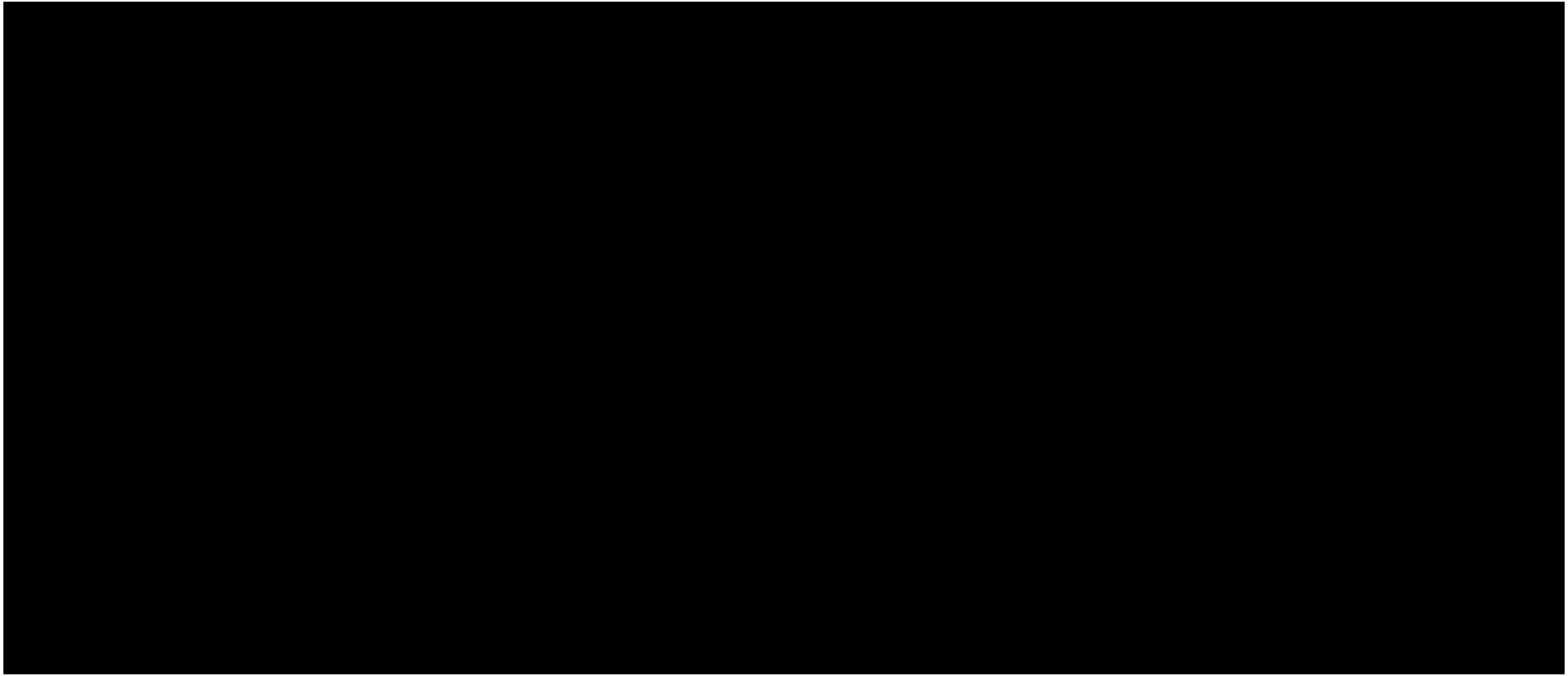
Trends: S-Curve

LOGISTIC GROWTH

Growth rate decreases as population reaches maximum capacity

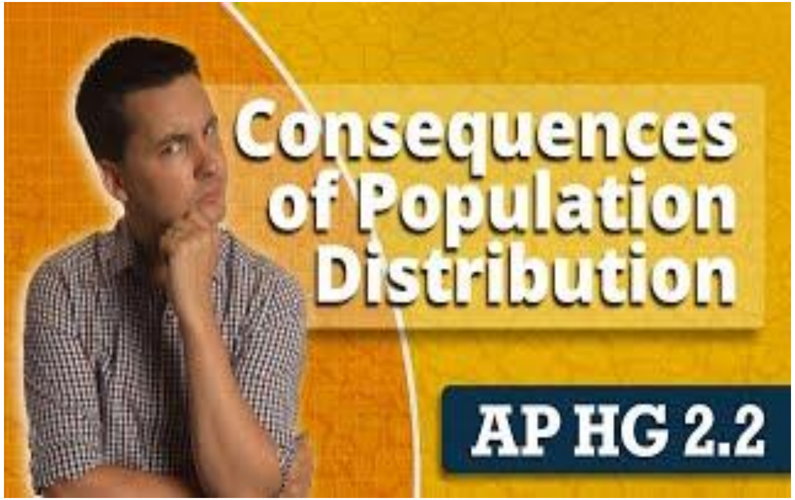


Carrying Capacity Defined: Watch Video to record notes.



Carrying Capacity

Review Videos: Watch & add/clarify notes as needed



AP Classroom Questions

Go to AP Classroom: Link Here

Go to Assignment: 2.1 & 2.2 Progress Check and complete questions.

Population Trends

Enrichment Activity

Use the Letter Booklet 2022 World Population to explore population trends. You can use the Population Trend Worksheet as a guide to your exploration. It is helpful to know general regional population trends.